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Objective

Transform a CPA secure encryption scheme $P.\Sigma$ into a CCA secure encryption scheme $FO.\Sigma$

1 Fujisaki Okamoto transform

1.1 Given

- Ciphertext space $\mathfrak{C}$, message space $\mathfrak{M}$.
- Public key encryption scheme $P.\Sigma$
- Symmetric key encryption scheme $S.\Sigma$
- Hash functions H_1, H_2 .
- Message 'm'.

1.2 $FO.KGen(1^\lambda)$

- $(pk, sk) \xleftarrow{R} P.\Sigma.KGen(1^\lambda)$

1.3 $FO.Enc(pk, m) \rightarrow c_1, c_2$

- $x \xleftarrow{R} \{0, 1\}^* = \mathfrak{M}$ (message space)
- $c_1 \xleftarrow{R} P.\Sigma.ENC_{pk}(x, H_1(x))$
- $k = H_2(x)$, hashes are deterministic
- $c_2 \xleftarrow{R} S.\Sigma.ENC_k(m)$,

1.4 $FO.Dec(sk, c_1, c_2) \rightarrow m$

- $x = P.\Sigma.DEC_{sk}(c_1)$
- $k = H_2(x)$
- $m = S.\Sigma.DEC_k(c_2)$
- Check if $c_1 = P.\Sigma.ENC_{pk}(x, H_1(x))$, Accept if yes, Abort otherwise

2 Questions

- Why is $H_1(x)$ used in P.Σ.ENC?
(ANS.) $H_1(x)$ is here for making P.Σ.ENC probabilistic and for a check by the receiver.
- c_2 should not be randomized, there's nothing to randomize, or is it?
- If $H_1(x)$ is not a concatenated message, then is it shared by the both parties?
(ANS.) Well, it is concatenated as in AES, the IV is concatenated with the ciphertext

